NEPHLUX: Machine Learning–Based Albuminuria Risk Stratification from PhilHealth YAKAP Labs for Early Kidney Damage

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Abstract. *Hemodialysis was PhilHealth's top-paid medical procedure in 2024, at up to PHP 990,600 per patient per year. Early kidney damage is found with a urine albumin-creatinine ratio (uACR) test, which PhilHealth's YAKAP primary care benefit does not cover. NEPHLUX ranks adults for that test using what a YAKAP clinic already collects; each pass mark was written first. Step 2 (YAKAP blood tests, blood pressure and history) was trained on U.S. NHANES 2011–2016 and tested once on NHANES 2017–March 2020 (6,655 adults): AUROC 0.757 (95% CI 0.737–0.775), calibration slope 0.95, and 32 more cases than the rule "refer everyone with diabetes or high blood pressure" at the same 49% referred (95% CI +12 to +51). NHANES has no urine dipstick, but YAKAP covers urinalysis. Step 2U adds the dipstick to Step 2 and was learned on Korean KNHANES 2021–2024 (20,465 adults). Scored once on 11,047 Korean adults from KNHANES 2019–2020 (912 with uACR ≥30 mg/g), it reached an AUROC of 0.917 (95% CI 0.909–0.926) and a calibration slope of 1.02, and, referring the same 39.4% as "diabetes or high blood pressure or dipstick protein 1+", found 846 cases against 752 (+94; 95% CI +72 to +118). Without the dipstick, Step 2 found fewer cases than that rule (−54; 95% CI −78 to −27). A questionnaire with a measured blood pressure and waist found 17 more cases than the plain rule (95% CI +1 to +35); questions alone did not, so the home check is for awareness only. A web app set in Cagayan de Oro City (80 barangays, 42 YAKAP clinics), with the largest share of CKD admissions at its tertiary government hospital, serves residents, YAKAP clinics and the City Health Office, which sees counts only. The models were tested on American and Korean adults, not Filipinos; Philippine testing is next.*

Keywords: *chronic kidney disease, albuminuria, screening, PhilHealth YAKAP, prediction model, NHANES, KNHANES, urinalysis, primary care, decision support, Cagayan de Oro*

I. INTRODUCTION

a. Background

Chronic kidney disease (CKD) is defined by reduced kidney function or by markers of kidney damage, most commonly albumin in the urine, lasting at least three months. Albuminuria is staged by the urine albumin-creatinine ratio (uACR), with 30 mg/g or more counted as raised (Kidney Disease: Improving Global Outcomes [KDIGO], 2024). Higher albuminuria predicts kidney failure at every level of filtration (Gansevoort et al., 2011), and it is treatable: in type 2 diabetes with nephropathy, losartan cut the risk of end-stage kidney disease by 28% (Brenner et al., 2001). Losartan and enalapril are already in the YAKAP drug list (Philippine Health Insurance Corporation [PhilHealth], 2024a).

In the Philippines the end stage is expensive. Hemodialysis was PhilHealth's top-paid medical procedure in fiscal year 2024 (PhilHealth, 2025). Since October 2024 PhilHealth pays PHP 6,350 per session for up to 156 sessions a year, or PHP 990,600 per patient (PhilHealth, 2024b). The same circular quotes the National Kidney and Transplant Institute: about one Filipino develops chronic renal failure every hour.

Early diagnosis is rare. The Philippine Society of Nephrology said in 2024 that fewer than 10% of Filipino adults had been diagnosed with CKD in 2021, that screening blood and urine tests were not in PhilHealth packages, and urged PhilHealth and local governments to cover them (Orejas, 2024).

Local data are thinner still. No published source gives CKD prevalence, dialysis patients or kidney deaths by Philippine city or barangay; national cause-of-death releases stop at the national level, and the Philippine Renal Disease Registry does not publish city figures. The nearest local evidence is a hospital study in Northern Mindanao: of 32,378 adult admissions at Northern Mindanao Medical Center (May 2024 to May 2025), 1,207 had CKD, 580 of them from Cagayan de Oro City, more than from any other city or municipality. 82.4% arrived at stage 5 and 98.4% were already on dialysis; hypertension (35.7%) and diabetes (34.8%) were the leading causes (Mambering & Espiga-Comonsad, 2026). These are admissions near one hospital, not a city rate, but they show the pattern this project targets: kidney disease found when dialysis is the only option left.

b. Statement of the Problem

PhilHealth's Yaman ng Kalusugan Program (YAKAP), formerly Konsulta, gives registered members primary care that is free at public YAKAP clinics, with a co-payment capped at PHP 900 a year at private ones (PhilHealth, 2026a), and a set list of tests: complete blood count, lipid profile, fasting blood sugar, oral glucose tolerance test, HbA1c, creatinine, chest X-ray, sputum microscopy, ECG, mammogram, ultrasound, urinalysis, Pap smear, fecalysis and fecal occult blood test (PhilHealth, 2024a). The uACR test is not on the list.

Creatinine, the kidney test YAKAP does cover, finds kidney damage only once filtration falls. In our NHANES test set, 85% of adults with uACR ≥30 mg/g had an estimated glomerular filtration rate (eGFR) of 60 or more, a normal-looking creatinine result (survey-weighted: 89%). The usual shortcut, testing people with diabetes or high blood pressure, missed 24% of cases, who had neither.

Problem statement. A YAKAP doctor cannot order a uACR test for everyone, but has no evidence-based way to decide who needs one from the tests YAKAP already pays for.

Second, a local gap. A City Health Office that wants to target kidney screening has no barangay-level picture of who is at risk or where screening is reaching, because those numbers are not collected anywhere.

c. Objectives

The general objective is to build and test a two-step kidney screen that uses only what YAKAP clinics already collect, and to report honestly whether it beats simple rules. Specifically, to:

1. fix a pass mark before training: beat the rule "refer everyone with diabetes or high blood pressure" at the same number of referrals, and beat the AUROC of about 0.70 reported for MERWACS (Yoo et al., 2026);
2. build step 1, a no-lab questionnaire a resident can complete, step 2, a model on YAKAP-covered labs, blood pressure and history, and step 2U, which adds the YAKAP urine dipstick;
3. train on NHANES 2011–2016 and test once on NHANES 2017–March 2020, then test on Korean KNHANES, which has a dipstick, reporting discrimination, calibration, net benefit, Asian-subgroup results and reasons for each prediction;
4. estimate the cost per kidney case found using PhilHealth's published rates;
5. build a web app for residents, YAKAP clinics and the City Health Office of Cagayan de Oro City, the city with the largest share of CKD admissions at its tertiary government hospital (Mambering & Espiga-Comonsad, 2026), and turn each clinic's referrals into anonymous barangay counts of screening reach;
6. build an in-app help assistant that answers questions about the app, YAKAP and kidney disease without giving medical advice, and test it against a pass mark written before its answers; and
7. plan the outside checks the system still needs.

d. Scope and Boundaries

- **Screening, not diagnosis.** The output is a referral suggestion for a uACR test. Albuminuria is confirmed only by repeat uACR, and CKD only over three months (KDIGO, 2024).
- **Adults 20 and older without known kidney disease.** People already told they have kidney disease or on dialysis are outside the target, since they need care, not screening.
- **U.S. and Korean data.** No Philippine dataset with uACR and YAKAP labs was available to the team. Every performance number is from NHANES or KNHANES and must be re-checked in Filipino patients.
- **Cagayan de Oro City pilot design.** Barangay boundaries (PSGC 2023) and populations (2020 census) are real, and the 42 YAKAP clinics are PhilHealth's accredited list as of September 30, 2026 (PhilHealth, 2026b; map pins approximate). Dashboard records and the story patient are labelled samples.
- **Screening reach, not prevalence.** Barangay counts cover only adults who reach a YAKAP clinic. They show where screening is reaching and where referrals cluster, not how common CKD is in the barangay.
- **No link to PhilHealth systems.** The app files no claims and reads no PhilHealth records.

II. REVIEW OF RELATED LITERATURE AND SYSTEMS

a. Albuminuria, Filtration and Kidney Failure

KDIGO (2024) recommends testing both eGFR and uACR in people at risk of CKD and classifying CKD by both. In pooled general-population and high-risk cohorts, kidney failure risk rose steeply with albuminuria even when eGFR was normal (Gansevoort et al., 2011). eGFR is estimated here with the race-free 2021 CKD-EPI creatinine equation (Inker et al., 2021), the equation current guidance prefers.

b. Kidney Care in PhilHealth and YAKAP

PhilHealth covers the end of the disease well: 156 hemodialysis sessions a year at PHP 6,350 each (PhilHealth, 2024b), with peritoneal dialysis and transplant packages besides (PhilHealth, 2025). The front end is thinner. YAKAP covers creatinine and urinalysis but not uACR (PhilHealth, 2024a), and nephrologists have asked for screening tests to be covered (Orejas, 2024). A YAKAP urinalysis includes a dipstick protein reading, which is far less sensitive than uACR for early albuminuria (KDIGO, 2024).

c. Risk Models for CKD Screening

MERWACS (Yoo et al., 2026) is the closest prior work. It used 12 non-laboratory inputs (demographics, body measures, history), trained XGBoost on NHANES, and validated externally on the Korea NHANES (KNHANES). Its AUROC was 0.68–0.70 internally and 0.71–0.73 externally. Its outcome is a composite of uACR ≥30 mg/g or eGFR below the age- and sex-specific 2.5th percentile, so it predicts albuminuria and low filtration together. MERWACS deliberately avoids labs; our question is how much the labs YAKAP already pays for add.

d. Lab-Only Screening in YAKAP Clinics

e. Related Systems and Research Gap

Gap. There is evidence that albuminuria predicts kidney failure, a primary care benefit that already pays for most of the inputs, and no-lab models with AUROC near 0.70, but no tested tool that turns YAKAP-covered labs into a uACR referral decision and states, before testing, what it must beat. Nor is there any city- or barangay-level measure of kidney risk: the only local map is of hospital admissions (Mambering & Espiga-Comonsad, 2026).

III. METHODOLOGY

a. Design and Pre-Registered Pass Mark

This is a prediction model development study with temporal validation, reported with TRIPOD+AI (Collins et al., 2024). Before any data were downloaded, we wrote the pass mark to a file (PASS_MARK.md, dated October 17, 2026): on the 2017–2020 test set, step 2 must have an AUROC whose 95% confidence interval lies above 0.70, and must find more cases than the diabetes-or-high-BP rule at the same number of referrals, with the bootstrap interval of the difference excluding zero. A calibration slope between 0.8 and 1.2 was also required. Gradient-boosted trees would replace logistic regression only if they improved cross-validated AUROC on the training cycles by at least 0.01. Failures are reported as failures.

b. Data Sources

Table 1. *Data sources (all downloaded as raw XPT files from the CDC; checksums in the repository)*

Data	NHANES files	Use
Demographics, survey weights	DEMO	Age, sex, race/ethnicity, pregnancy, weights
Urine albumin, creatinine	ALB_CR	Outcome: uACR ≥30 mg/g
Serum creatinine	BIOPRO	eGFR, CKD-EPI 2021 (YAKAP: creatinine)
HbA1c	GHB	YAKAP: HbA1c
Total and HDL cholesterol	TCHOL, HDL	YAKAP: lipid profile
Hemoglobin, WBC, platelets	CBC	YAKAP: CBC with platelets
Fasting glucose	GLU	Not used (fasting subsample only)
Blood pressure	BPX (2011–2016), BPXO (2017–2020)	Measured BP
Body measures	BMX	Body-mass index
Questionnaires	DIQ, BPQ, SMQ, MCQ, KIQ_U	History; exclusion of known kidney disease
Hemodialysis rate	PhilHealth Circular 2024-0023	Cost per dialysis-year
Cagayan de Oro barangays	2020 census (Philippine Statistics Authority [PSA], 2021; Abdulsamad, 2026); PSGC 2023 boundaries (faeldon, 2024)	City dashboard denominators

Sixty files were used: 15 per cycle for 2011–2012, 2013–2014, 2015–2016 and the 2017–March 2020 pre-pandemic files (Centers for Disease Control and Prevention [CDC], n.d.; Stierman et al., 2021). We checked which YAKAP tests NHANES contains (Table 2). NHANES has no dipstick urinalysis, ECG, chest X-ray or imaging in these cycles, so those YAKAP tests could not be modelled. The Korean tests used KNHANES 2011–2014 and 2019–2024 (Korea Disease Control and Prevention Agency [KDCA], n.d.-a) under its data-use terms; the raw files are not shared.

Table 2. *YAKAP-covered tests (PhilHealth, 2024a) and their use in the model*

YAKAP test	In NHANES 2011–2020?	In model
CBC with platelets	Yes	Hemoglobin, WBC, platelets
Lipid profile	Total and HDL all; LDL and TG fasting only	Total, HDL
FBS / OGTT	Fasting subsample / not all cycles	No (HbA1c used)
HbA1c	Yes	Yes
Creatinine	Yes (serum)	Yes, as eGFR
Urinalysis (dipstick)	No	No (limitation)
ECG, chest X-ray, ultrasound	No	No
uACR (outcome)	Yes	Not a YAKAP test

c. Participants and Outcome

Table 3 shows how the cohorts were formed. Adults aged 20 or older were kept; pregnant women were removed because pregnancy raises albumin excretion; people who reported being told they had weak or failing kidneys, or dialysis in the past year, were removed because they need care, not screening. Records missing any step-2 input were removed (complete-case analysis); missing diastolic pressure, creatinine and blood-pressure exams were the main causes. The outcome is a single spot uACR ≥30 mg/g.

Table 3. *Participant flow*

Step	Train 2011–16	Test 2017–20
All NHANES participants in DEMO	29,902	15,560
Age 20+	17,048	9,232
Not pregnant	16,856	9,145
Has urine albumin-creatinine ratio	15,827	8,234
Answered diabetes and BP questions	15,827	8,234
No known kidney disease or dialysis (screening target)	15,274	7,912
Complete BMI, BP and YAKAP labs (complete-case)	13,840	6,655

d. Predictors

Step 1 (no labs): age, age squared, sex, told has diabetes, told has high blood pressure, takes BP medicine, body-mass index, current smoking, parent or sibling with diabetes, and past heart attack, stroke or heart failure. **Step 2** adds mean systolic and diastolic pressure, HbA1c, eGFR (with an extra term for each 10 units below 90), total and HDL cholesterol, hemoglobin, white cell count and platelets. "Don't know" answers were treated as no. Continuous values were clipped to plausible ranges. Race was not used.

e. Models

Both steps use L2-penalised logistic regression (scikit-learn; Pedregosa et al., 2011) on standardised inputs, without re-weighting. In 5-fold cross-validation on the training cycles, step-2 AUROC was 0.757 for logistic regression and 0.760 for histogram gradient-boosted trees, a gain of 0.004, below the 0.01 bar, so logistic regression was kept. (XGBoost itself could not load on the team's machine; scikit-learn's equivalent boosted-tree method was used for this check.) Its coefficients are exported to the browser, where the same formula runs.

Risk levels were set on the training data only. **High** (refer for uACR) is the top share of risk equal to the share the rule refers in training; **moderate** is the level above which 90% of training cases fall; step 1's "see a YAKAP doctor" level is set the same way.

f. Evaluation

All metrics are on the untouched 2017–2020 test set. AUROC intervals come from 1,000 bootstrap resamples (fixed seed). Calibration is assessed by calibration slope and calibration-in-the-large (Van Calster et al., 2019). Net benefit is computed across referral thresholds (Vickers & Elkin, 2006). For the matched comparison, each model refers the same number of people as the rule, its highest-risk ones. Reasons are SHAP values (Lundberg & Lee, 2017); for a logistic model with independent inputs these are exactly coefficient × (value − training mean), which we checked against the shap package (maximum difference 0). Prevalence uses NHANES mobile-exam weights; model performance is unweighted, with weighted AUROC as a sensitivity check.

g. Korean Data and Step 2U

KNHANES, unlike NHANES, measures a urine dipstick and uACR in the same adults. Dipstick codes follow the KDCA user guidebook (KDCA, n.d.-b): protein code 2 is 1+. Three pass marks were written before the data they judge were opened. First, the frozen NHANES models were run once on KNHANES 2021–2024 (20,465 adults), with nothing refitted. Second, Step 2U, the Step 2 inputs plus dipstick protein (trace, 1+, 2+ or more), blood, glucose and specific gravity, was fitted with the same penalised logistic regression on KNHANES 2021–2024; earlier years use a different dipstick coding and were not used. Third, Step 1 v3, the questionnaire with and without a measured blood pressure and waist, was refitted on KNHANES 2011–2014 and 2021–2024 (39,707 adults). Step 2U and Step 1 v3 were then frozen and scored once, together, on KNHANES 2019–2020. Step 2U's main comparator is what a clinic would do with the dipstick anyway: refer anyone told of diabetes or high blood pressure, on BP medicine, with BP 140/90 or above, or with dipstick protein 1+.

h. Cost Analysis

Step-2 labs are already in the YAKAP benefit, so the new cost is the uACR test. We use PHP 600 for a public or bulk purchase (assumed, range 300–1,200; we found no PhilHealth rate for it) and PHP 1,500–3,500, the retail price at hospitals and diagnostic centers (Baclig, 2026). One dialysis-year is PHP 990,600 (PhilHealth, 2024b). We report the cost per case found and the dialysis-years that must be avoided per 1,000 adults screened to repay the tests. The PHP 900 yearly co-payment a private YAKAP clinic may charge (PhilHealth, 2026a) is not counted. We do not claim dialysis avoided, because NHANES has no follow-up.

i. Ethics and Privacy

- NHANES data are public and de-identified; no Filipino patient data were used.
- The resident check stores nothing: the server scores the answers and keeps no record or log of them.
- The help assistant, Nephra, stores nothing. When its optional AI model is switched on, the question is sent to that service to reword an approved answer, and the chat says so.
- Clinic records use initials only and are demo data. A deployment would make the City Health Office the controller under the Data Privacy Act of 2012.
- The city view shows counts only, with cells of 1 to 4 suppressed.
- The doctor decides every referral; an override requires a written reason.

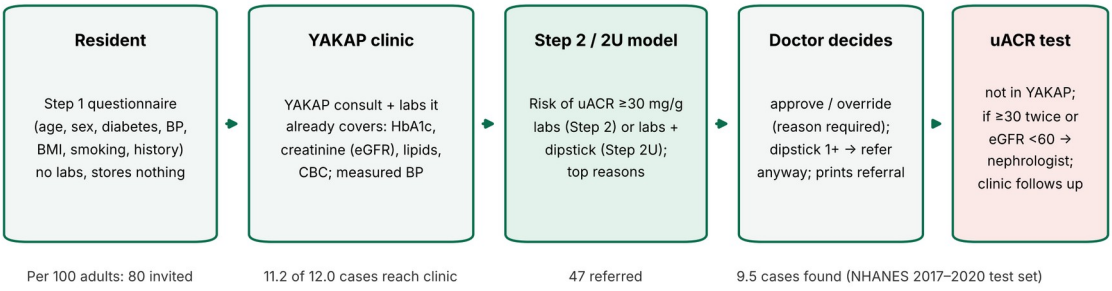


Figure 1. The two-step pathway. *Numbers under each box are Step 2 results per 100 adults in the NHANES 2017–2020 test set, using thresholds fixed on training data. Step 1 sends most adults on, because it is weak; step 2 does the sorting.*

IV. SYSTEM DESIGN

a. The Web App

NEPHLUX is a web app: a React front end (Tailwind CSS, Zustand) and a NestJS API on PostgreSQL. Its API runs Step 2U from the same coefficients, and an automated test checks that it equals the Python model on 300 generated patients. The screens in Figure 2 come from the single-file design prototype. Residents start the check with no account; YAKAP clinic and City Health Office staff sign in to their own screens (demo accounts, labelled not secure). A test checks that the browser gives the same risk as Python for 25 test-set participants (difference below 10^{-9}).

- **Residents.** Ten questions in English, Bisaya or Tagalog (the Bisaya and Tagalog text are drafts awaiting review by native speakers). The result says "higher" or "lower" chance, shows what raised it,

always ends with "get your YAKAP check-up" (free at public clinics, up to PHP 900 a year at private ones) and the nearest of the city's 42 YAKAP clinics, and gives a 9-character code that carries the answers to the clinic without saving them anywhere, and is never called a diagnosis.

- **YAKAP clinic (nurse or doctor).** A patient queue; a nurse or the doctor loads the resident's code, confirms each answer and enters labs in SI or conventional units. Missing labs block the estimate. The result shows risk, level, eGFR, the top reasons and a warning when eGFR is below 60; the doctor approves, overrides (a reason is required) or chooses no referral, and prints a uACR referral slip. A urine dipstick protein of 1+ or more means refer for uACR whatever the score; with a protein result entered, the app runs Step 2U (refer at 2.55% or more, a cut fixed on the Korean learning data), and without one it runs Step 2. A Follow-up tab lists the clinic's own referrals with no uACR result after 14 days.
- **City Health Office.** A map of Cagayan de Oro's 80 barangays with the 42 YAKAP clinics, labelled "screening reach, not CKD prevalence"; anonymous counts per 1,000 residents; a referral funnel; outreach priorities for large barangays with low reach; and a slider for the yearly uACR test budget with its cost per case.
- **Evidence.** A model card showing the pass mark and measured results.
- **Nephra, the help assistant.** A help chat on every screen that answers from 78 approved answers about the app, YAKAP and kidney basics and stores nothing. An optional AI model may reword the chosen answer in English, Filipino or Bisaya, but cannot add facts, calculate or diagnose; without one, the approved text is shown. Danger signs get the fixed emergency answer; diagnosis and medicine questions go to the doctor.

b. From Referrals to Barangay Screening Counts

NEPHLUX does not wait for a survey to fill the local data gap. Each time a YAKAP clinic scores a patient, the clinic keeps the record and sends the City Health Office only counts by barangay: screened, high risk, referred, uACR completed and uACR raised, with cells of 1 to 4 hidden. Over a year these counts become a barangay-level picture of where kidney screening reaches, not prevalence, because it covers only adults who reach a YAKAP clinic. Barangays with few people screened show where outreach should go next.

c. Reproducibility

One script (analysis/run_all.py) downloads nothing by itself, reads the 60 XPT files, and regenerates every number, table and figure in this paper and the browser model. A shell script fetches the files from the CDC and a checksum file confirms them. Seeds are fixed. Eleven automated checks (tests/engine.test.js) passed on October 17, 2026. The design prototype is one file (NEPHLUX_Prototype_v7.html). Its 39 checks (tests/prototype_v7.test.js, passed October 23, 2026) confirm that its model and engine are identical to the tested build and give the same scores on 5,000 random adults, that the browser Step 2U equals the Python model on 300 random patients, that the city data match the census and PhilHealth's list, that every headline number matches the evidence file, and that the page contains no team names, no forbidden claims and no storage or network calls in the resident check or in Nephra.

Table 4. *Users and what they get*

User	Input	Output
Resident	10 questions, no labs	Awareness result, clinic code, nearest YAKAP clinics
YAKAP doctor	Labs YAKAP covers, BP, history	Risk, reasons, referral slip, follow-up list
City Health Office	Counts only	Barangay map, funnel, test budget, cost planning

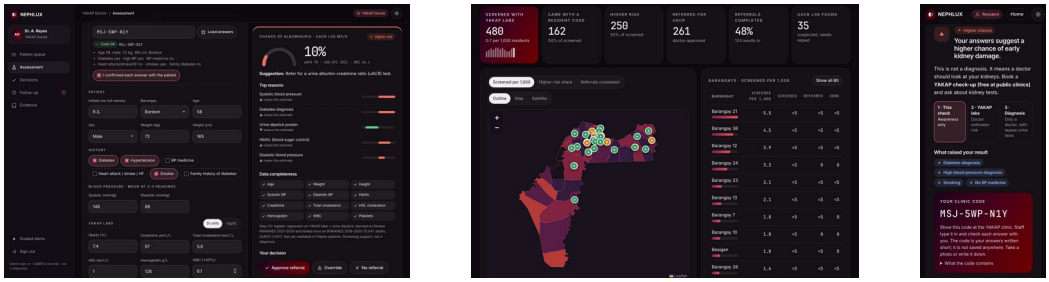


Figure 2. *The design prototype (v7). Left: YAKAP doctor's Step 2U result for the SAMPLE story patient, with top reasons, data completeness and the referral decision. Middle: City Health Office view of Cagayan de Oro's 80 barangays and 42 YAKAP clinics (SAMPLE records; pins approximate). Right: resident awareness result and clinic code at phone width.*

V. RESULTS

a. Participants

The training set had 13,840 adults (1,545 with uACR ≥30 mg/g, 11.2%); the test set had 6,655 (797, 12.0%). Weighted to the U.S. population, test-set prevalence was 9.2%. Of the test-set adults, 14.4% reported diabetes and 37.3% high blood pressure. Albuminuria was present in 18.7% of those with diabetes or high blood pressure and in 5.5% of those with neither.

b. Pass Mark and Success Criteria

Table 5. *Pre-registered criteria and measured results (test set 2017–2020)*

Criterion	Target	Result
Step 2 AUROC vs MERWACS	95% CI above 0.70	0.757 (95% CI 0.737–0.775) — Pass

Criterion	Target	Result
Step 2 cases found vs rule, same referrals	More, CI excludes 0	+32 (+12 to +51) — Pass
Step 2 calibration slope	0.8–1.2	0.95 (0.87–1.02) — Pass
Step 1 AUROC	95% CI above 0.70	0.718 (95% CI 0.699–0.737) — Fail
Step 1 cases found vs rule	More, CI excludes 0	-1 (-18 to +19) — Fail

Step 2 passed all three criteria. Step 1 failed both of its criteria: without labs, it does no better than asking about diabetes and blood pressure. Its role in the app is therefore to bring people to the YAKAP check-up, not to sort them.

c. Discrimination, Calibration and Net Benefit

Table 6. *Discrimination on the test set (AUROC, 95% bootstrap CI)*

Approach	AUROC
Step 2: YAKAP labs + BP + history	0.757 (95% CI 0.737–0.775)
Step 2, gradient-boosted trees (not chosen)	0.766 (95% CI 0.748–0.784)
Step 1: questionnaire	0.718 (95% CI 0.699–0.737)
Rule: diabetes or high BP	0.656 (95% CI 0.639–0.671)
Age alone	0.644 (95% CI 0.623–0.663)
eGFR below 60 alone	0.553 (95% CI 0.540–0.565)
MERWACS, reported (composite outcome)	0.68–0.70 internal; 0.71–0.73 KNHANES

Labs added 0.039 AUROC over the questionnaire (95% CI 0.026–0.052). Survey-weighted AUROC was 0.735 for step 2 and 0.711 for step 1. Mean predicted risk was 12.0% against an observed 12.0% (calibration-in-the-large -0.008); Brier score 0.092. Step 2 had the highest net benefit of all strategies at every threshold from 5% to 50% (Figure 3B).

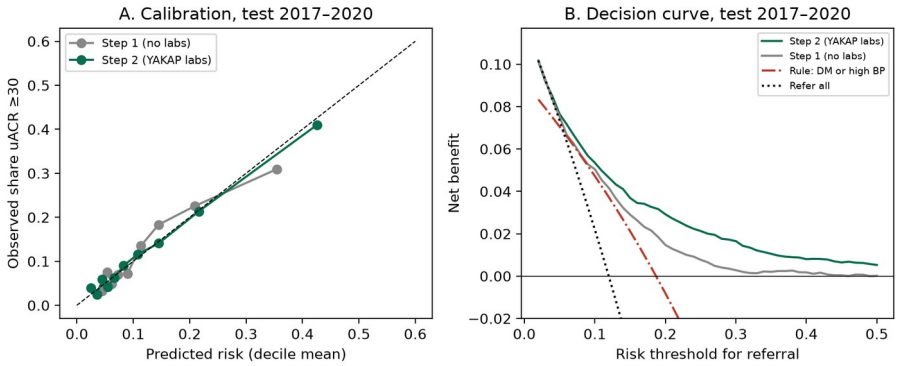


Figure 3. Test-set calibration and decision curve. *A: observed against predicted risk by decile. B: net benefit across referral thresholds for step 2, step 1, the diabetes-or-high-BP rule and testing everyone (NHANES 2017–March 2020).*

d. Same Number of Referrals

Table 7. *Test set, each approach referring 49.0% of adults (3,260 of 6,655)*

Approach	Cases found	Sensitivity	PPV
Rule: diabetes or high BP	609 / 797	76.4%	18.7%
Step 1: questionnaire	608 / 797	76.3%	18.7%
Step 2: YAKAP labs	641 / 797	80.4%	19.7%

Per 1,000 adults, step 2 finds about 96 cases with 490 uACR tests; the rule finds 92 with the same number.

e. The Two-Step Pathway per 100 Adults

With thresholds fixed on training data, step 1 invited 80 of every 100 adults to a YAKAP check-up, and those invited included 11.2 of the 12.0 cases (93%). Step 2 then referred 46.8 for uACR and found 9.5 cases, 79% of all cases, with one case per 5.0 tests. The rule, applied to everyone, found 9.2 per 100 with 49.0 tests.

f. Which YAKAP Tests Add Value

Table 8. *Nested models on the test set (refitted on training data)*

Model	AUROC (95% CI)	Cases*
questionnaire only	0.718 (0.698–0.735)	608
+ measured BP	0.727 (0.707–0.746)	613
+ HbA1c	0.742 (0.720–0.762)	625
+ creatinine (eGFR)	0.750 (0.729–0.770)	634
+ lipid profile	0.750 (0.729–0.770)	635
+ CBC (full step 2)	0.757 (0.736–0.775)	641

Cases column: cases found when referring the same share as the rule. HbA1c and creatinine carried most of the gain; the lipid profile added nothing measurable; the CBC added a little, mainly through hemoglobin.

g. Subgroups and the Korean Outside Test

In 789 Asian participants (88 cases), step-2 AUROC was 0.741 (95% CI 0.689–0.792), step 1 0.719 (95% CI 0.664–0.774) and the rule 0.663 (95% CI 0.614–0.712). At the rule's referral share, step 2 found 64 cases against the rule's 62; calibration slope was 0.98 (0.73–1.24), with risk slightly under-predicted (mean predicted 9.7%, observed 11.2%). The interval's lower bound is below 0.70, so for Asian adults the pass mark is not shown to hold. AUROC was similar across the other race/ethnicity groups (0.73–0.78), lower in women (0.71) than men (0.80), and lowest in adults aged 20–44 (0.68).

In 20,465 Korean adults (1,394 cases, 6.8%), the frozen step-2 model reached an AUROC of 0.784 (95% CI 0.772–0.796) against 0.699 (0.688–0.710) for the rule. Referring the same 39.6% as the rule, it found 1,097 cases

against 1,069 (+28; 95% CI +5 to +53). It failed calibration: slope 1.26 (1.19–1.32), with mean predicted risk 9.1% against 6.8% observed. Recalibrated on a random half, the slope on the other half was 1.03 (0.95–1.10), reported separately from the frozen result.

h. Step 2U on a Clean Korean Test

On KNHANES 2019–2020 (11,047 adults, 912 cases, 8.3%), Step 2U passed all four of its criteria. Its AUROC was 0.917 (95% CI 0.909–0.926), against 0.786 (0.770–0.802) for the same Korean model without the dipstick (difference 0.131; 0.116–0.147), 0.779 (0.763–0.795) for the frozen NHANES Step 2 and 0.735 (0.721–0.748) for the rule with the dipstick. Its calibration slope was 1.02 and calibration-in-the-large −0.05. Referring the same 39.4% as the rule with the dipstick, Step 2U found 846 of the 912 cases (93%) against 752 (82%): +94 (95% CI +72 to +118). Among adults with neither diabetes nor high blood pressure (244 cases) it caught 80%, against 34% for the rule with the dipstick.

Without the dipstick, neither version of Step 2 beat a clinic that refers on a positive dipstick: the Korean refit found 35 fewer cases (95% CI −57 to −10) and the frozen NHANES Step 2 54 fewer (−78 to −27). Against the plain rule on this third test set, the frozen Step 2 found 21 more cases, with an interval that reaches below zero (−1 to +41). Their value is therefore in combination with the dipstick.

Step 1 v3 with a measured blood pressure and waist passed its pass mark: AUROC 0.778 (0.762–0.794), 17 more cases than the rule at the same 38.0% referred (95% CI +1 to +35), calibration slope 1.04. With questions only it reached 0.754 (0.735–0.769) and did not beat the rule (−8; −29 to +15), so the home check stays awareness only.

i. Reasons Behind Each Score

Across the test set, the inputs with the largest average SHAP values were eGFR (both terms), systolic pressure, HbA1c and age. eGFR acts in both directions: low values raise risk, and so do very high values, consistent with hyperfiltration.

j. Cost per Case Found

Table 9. *Per 1,000 adults screened (test-set rates; uACR PHP 600, public or bulk (assumed range 300–1,200); retail PHP 1,500–3,500)*

Strategy	uACR tests	Cases	PHP per case found (PHP 600; retail)	Break-even dialysis-years
Test everyone	1000	120	5,010 retail 12,525–29,225	0.61
Rule: diabetes or high BP	490	92	3,212 retail 8,030–18,737	0.30
Step 2, same referrals	490	96	3,051 retail 7,628–17,798	0.30
Two-step pathway	468	95	2,971 retail 7,428–17,331	0.28

Testing everyone finds all 120 cases per 1,000 at PHP 5,010 per case. The two-step pathway finds 95 at PHP 2,971 per case and misses 25. At PHP 600 the tests cost 0.61 of one dialysis-year for testing everyone and 0.28 for the two-step pathway; at retail prices, 1.51 to 3.53 and 0.71 to 1.65. With Step 2U on the Korean 2019–2020 test, 50 uACR tests per 1,000 adults found 39.7 cases against 9.1 for the rule with the dipstick (+30.6; 95% CI +28.6 to +33.0), about PHP 755 against PHP 3,281 per case at PHP 600 and PHP 1,887–4,404 against PHP 8,203–19,141 at retail prices; this is the budget view the City Health Office screen shows.

k. Testing of the App and the Help Assistant

The v7 app passed its 39 automated checks (no storage or network calls in the resident check or Nephra, no forbidden claims or team names), and an accessibility scan of the v6 screens found no serious issues on 9 views.

Nephra v1.2 was scored once on held-out questions against a pass mark written before its answer bank (PASS_MARK_ASSISTANT.md). It gave no unsafe answer and caught every danger phrasing, but gave the right answer first time to only 38 of 62 in-scope questions (61%; target 85%), 48 of 62 (77%) within one "did you mean" tap. These scores are for the answer bank alone; the optional AI rewording was added later and has not been scored. Nephra is labelled a demo assistant until it is tested on questions written by residents and clinic staff.

VI. DISCUSSION

a. What the Results Mean

YAKAP already pays for the tests that carry the signal, but they work only when read together. On U.S. data the blood tests beat the diabetes-or-high-BP rule by a modest margin, about 5 more cases per 1,000 adults, and on the later Korean test that margin was not clearly above zero. The urine dipstick changes the picture: a clinic that reads the dipstick with HbA1c, creatinine and blood pressure found 94 more cases among 11,047 adults than one that refers everyone with diabetes, high blood pressure or a positive dipstick, for the same number of uACR tests. The honest claim is therefore not that NEPHLUX beats clinics with blood tests alone, but that it turns two tests a YAKAP clinic already has, the dipstick and the blood panel, into one ranking for the test YAKAP does not pay for. Without labs, screening does no better than asking two questions, so the home check is judged by how many people it brings to a clinic.

As a source of local data, NEPHLUX is a cheap first step, not a survey, to be checked later against a random-sample urine survey or the hospital's admissions by barangay.

b. Limitations

- **U.S. training data, no Philippine validation.** Filipinos differ in body size, diet, diabetes control and causes of kidney disease. NHANES Asian Americans are not Cagayan de Oro residents, and Asian-subgroup results were less certain.

- **Calibration did not transfer.** In Korean adults the NHANES model ranked well but over-predicted risk (slope 1.26); the Korean-trained Step 2U was well calibrated on its test years. Percentages need local recalibration before use in a new population.
- **Local burden not measured.** The case for Cagayan de Oro rests on admission counts at one hospital in the city (Mambering & Espiga-Comonsad, 2026), which favour nearby residents. Per-capita rates for the city and its neighbours were not computed.
- **Dipsticks read by a laboratory.** Step 2U was learned and tested on Korean adults whose strips were read by the survey's laboratory; a YAKAP strip read by eye may do worse. The blood and glucose strip codes are assumed; only the protein code was checked against the guidebook.
- **One uACR measurement.** Albuminuria is confirmed by repeat testing (KDIGO, 2024); some test-set "cases" are transient, which lowers apparent accuracy and inflates prevalence.
- **Blood pressure device changed.** Training cycles used auscultation and the test cycle an oscillometric device (Ostchega et al., 2021). Calibration held (slope 0.95), but Philippine clinic readings will differ again.
- **Complete-case analysis.** People missing any input were dropped; clinics with incomplete labs need a fallback.
- **Assumed uACR price and no outcome follow-up.** Costs per case depend on an assumed price (PHP 600) and rise in proportion at the published retail price (PHP 1,500–3,500); which strategy finds more cases per test does not change. Dialysis avoided was not estimated.
- **Demo only. Sign-in uses demo accounts and is not secure, records are SAMPLE data, clinic pins are approximate (8 addresses name no barangay), the Bisaya and Tagalog text has not been checked by native speakers, and no user has tested the app yet.**
- **Help assistant accuracy.** Nephra is safe but answered only 61% of new questions right first time, on questions the team wrote, and the AI rewording has not been scored.

c. Outside Checks

Three checks should come before any pilot:

- **Clinical review.** A YAKAP physician in Cagayan de Oro should review the clinic screen, the referral slip and the thresholds; a nephrologist, approached through the Philippine Society of Nephrology, should review the referral rules (repeat uACR, eGFR below 60) and the patient wording.
- **Walkthrough with one nurse or doctor.** One YAKAP nurse or doctor completes scripted tasks (enter labs and print a referral; find missed referrals on the Follow-up tab; read the barangay table), thinking aloud, and the problems found are fixed. It is a first look, not a usability study; it uses sample data and does not enter the clinic's work.
- **External validation in the Philippines.** The Korean tests (Sections V.g–h) show that the ranking transfers to an Asian national sample, that the NHANES calibration does not, and that the dipstick carries most of the gain. A Philippine dataset with uACR, for example from an LGU screening programme, is the real target.

d. Future Work

Validate on Philippine records with uACR and dipstick results; recalibrate for Filipino prevalence, as the Korean test showed is needed; test Step 2U on strips read by eye in YAKAP clinics; widen the walkthrough to a usability test with staff from several clinics and the City Health Office; connect lab entry to iClinicSys or the YAKAP e-form; and test Nephra again, with and without the AI rewording, on questions written by residents and clinic staff, scored once on a frozen build.

VII. CONCLUSION

Hemodialysis is PhilHealth's largest procedure bill, and the test that finds kidney damage early is the one YAKAP does not pay for. NEPHLUX uses the tests YAKAP does pay for to decide who should get that test. Against a pass mark written before training, the lab model passed: AUROC 0.757 (95% CI 0.737–0.775), well calibrated, and 32 more cases found than the usual rule for the same number of urine tests. Adding the YAKAP urine dipstick (Step 2U) raised the AUROC to 0.917 on a clean Korean test and found 94 more cases than a clinic using the dipstick on its own, for the same number of tests; the blood tests alone did not. The no-lab step failed and is used only to bring people to a YAKAP check-up. Set in Cagayan de Oro, where most CKD admissions at its tertiary government hospital already arrive at stage 5, the app also starts the barangay-level record of kidney screening that the city does not yet have. Its help assistant gave no unsafe answer but failed its accuracy bar, and the app labels it a demo. The NHANES models, the app and every NHANES number in this paper can be regenerated from public data with one script. What it has not done yet is meet a Filipino patient, and that is the next test.

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